Experiment-8

User Management and Role-Based Access Control for Library Book Issue Application

1.Aim:

To implement user management and Role-Based Access Control (RBAC) in the Library Book Issue application using Zoho Creator, ensuring that users can access forms, reports, and operations according to their assigned roles and permissions.

2. Procedure

1. Open the Library Book Issue application in Zoho Creator.
2. Identify the users who will access the application, such as the Administrator, Librarian, and Student.
3. Create or configure the required forms, including Student Details, Book Details, and Library Book Issue.
4. Configure user permissions and roles based on the responsibilities of each user.
5. Assign appropriate permissions to forms, reports, and application features.
6. Restrict sensitive operations, such as deleting records or modifying book details, to authorized users.
7. Test the application using different user roles to verify that access restrictions work correctly.

3. Member Access Design

Role-Based Access Control assigns permissions according to a user's role rather than granting every user unrestricted access.

Feature / Permission	Administrator	Librarian	Student
View student details	Full access	View and edit	Own details only, if configured
Add or modify book details	Yes	Yes	No
Issue books	Yes	Yes	No
View issue records	All records	All relevant records	Own records only, if configured
Return books and update status	Yes	Yes	No
Delete records	Yes	Restricted or no	No
Manage users and permissions	Yes	No	No

Note: The table represents the intended access design. Actual permissions must be configured in Zoho Creator. Students must not be able to view other students' records merely because a report is accessible to them.

4. Implementation

Step 1: Configure application users

- Open the Library Book Issue application.
- Navigate to the application's user and sharing settings.

- Add or invite the required users, where your Zoho Creator plan and account permissions allow it.
- Identify the Administrator, Librarian, and Student user groups.

Step 2: Configure roles and permissions

- Open the application's settings and locate the user permissions, roles, or sharing options available in your Creator interface.
- Assign administrative permissions to the Administrator.
- Allow the Librarian to manage book information, issue records, and returns.
- Restrict the Student to permitted viewing functions.
- Give each user only the permissions required for their work.

Step 3: Configure form access

Set suitable permissions for the following forms:

- Student Details: Restrict editing to authorized users.
- Book Details: Allow authorized staff to add and update book information.
- Library Book Issue: Allow authorized staff to create issue records and update return status.

Step 4: Configure report access

- Open the reports associated with the application.
- Restrict access to sensitive reports.
- Configure student-facing reports to show only the current student's records, using an appropriate ownership-based filter or server-side access control.
- Verify that users cannot bypass restrictions by opening a report or record directly.

Step 5: Save and test

Save the configuration and test the application using separate accounts assigned to each role. Do not assume that hiding a button alone provides security; the underlying data and operations must also be protected.

5. Process Diagram

Start

User opens the application

User Authentication

Verify the user's identity

Identify User Role

Administrator / Librarian / Student

Check Permissions

Verify access to the requested operation

Authorized

Allow permitted action

Unauthorized

Deny access

End

Record the outcome where appropriate

6. Verification and Testing

Test case	Expected result
Administrator opens the application	Can access authorized administrative features
Librarian issues a book	Issue record is created successfully

Student attempts to edit book details	Access is denied
Student attempts to delete an issue record	Access is denied
Student views their own issue history	Only their permitted records are displayed
Unauthorized user opens a restricted report	Access is denied
Librarian attempts to manage user permissions	Access is denied unless explicitly authorized

Verification method: Log in with separate test accounts for each role and perform the operations listed above. Record the actual result and compare it with the expected result.

7. Result

User management and Role-Based Access Control were designed for the Library Book Issue application in Zoho Creator. The access design separates Administrator, Librarian, and Student responsibilities and defines permissions for forms, reports, and book transactions. Verification confirms whether authorized users can perform their assigned operations and unauthorized users are prevented from accessing restricted functions.